Tidal Locking

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This paper briefly explains what tidal locking is and reviews the mechanism of tidal locking in a simple way. In addition, this paper presents the estimation of the time required for several astronomical bodies to be tidally locked. The effects of tidal locking on the spin period of the earth and on the distance between the earth and the moon are also estimated. The results generally agree with the observational data.

INTRODUCTION

Tidal locking is a phenomenon that exists in many two-body systems in the universe. For example, the moon is always facing the earth with the same side [1]. It is because the spin period and the orbital period of the moon are the same. This is a result of the tidal locking. For many other two-body systems in the universe, some astronomical bodies have already been tidally locked, while some are being tidally locked. It usually takes a very long time for astronomical bodies to be completely tidally locked and the time required for tidal locking depends on many factors.

For a two-body system in which the two astronomical bodies have different sizes, usually the larger astronomical body is considered as the primary body and the smaller body is considered to be the satellite. In the case where the two bodies have similar sizes, either body can be considered as the primary body and the other one is considered to be the satellite. Both bodies can be tidally locked in a two-body system, but the smaller body usually will be tidally locked in a shorter time [2].

Because the distances between the primary body and the different parts of the satellite are different, the gravitational force of the primary body that acts on different parts of the satellite is not uniform. As a result, tidal force arises and it distorts the shape of the satellite. This distortion makes the shape of the satellite become more like an ellipsoid [3]. When the spin period of the satellite is different from its orbital period, the semi-major axis of the ellipsoid would be misaligned with the line which connects the two centers of the bodies. This causes the primary body to have a gravitational torque on the satellite. This torque would change the spin rate of the satellite gradually and make the spin period of the satellite get close to its orbital period. When the two periods become equal, the semi-major axis of the satellite would align with the line connecting the two centers of the bodies and the satellite would be tidally locked [4].

Many factors affect the time it takes for the satellite to be tidally locked [5]. Some major factors include the distance between the two bodies, the masses of the two bodies, and the rigidity of the satellite. In this paper, the time required for tidal locking is roughly estimated for some bodies. Because the earth's spin period is increasing due to the tidal forces caused by the moon [6], the rate of change of the earth's spin period is calculated in this paper. Furthermore, tidal locking would change the radius of the satellite's orbit [7]. This change of radius is calculated for the Earth-Moon system. The results generally agree with the observational data.

THEORY

Tidal Force

The distances between the center of the primary body and the different parts of the satellite are different. In this paper, all bodies are considered to be uniform so their geometric center is the same as their center of mass. Let us assume the distance between two bodies' centers is r. For simplicity, all the points that lie on the plane AOB (as shown in the figure) can be assumed to be at the distance r from the center of the primary body.

If the distance between the primary body and the point i on the satellite is r_i, then we have:

$$r_i = r + \Delta r_i \tag{1}$$

According to the law of gravitation:

$$a_{g,i} = G\frac{M}{r_i^2} \tag{2}$$

Where $a_{g,i}$ is the gravitational acceleration at point i, G is the gravitational constant, and M is the mass of the primary body.

Then we have

$$a_{g,i} = G\frac{M}{(r+\Delta r_i)^2} = G\frac{M}{r^2}\frac{1}{\left(1+\frac{\Delta r_i}{r}\right)^2} \tag{3}$$

$$a_{g,i} \approx G\frac{M}{r^2}\left(1-2\frac{\Delta r_i}{r}\right) = G\frac{M}{r^2} - 2G\frac{M\Delta r_i}{r^3} \tag{4}$$

If we let

$$a_{g,center} = G\frac{M}{r^2}, a_{g,tidal,i} = -2G\frac{M\Delta r_i}{r^3} \tag{5}$$

Then it is clear that the tidal acceleration $a_{g,tidal,i}$ satisfies

$$a_{g,tidal,i}\begin{cases} > 0, when \Delta r_i < 0 \\ = 0, when \Delta r_i = 0 \\ < 0, when \Delta r_i > 0 \end{cases} \quad (6)$$

Because the tidal forces at different parts of the satellite are different, the shape of the satellite would be distorted and usually the resulting shape looks like an ellipsoid. This is shown in Fig. 1.

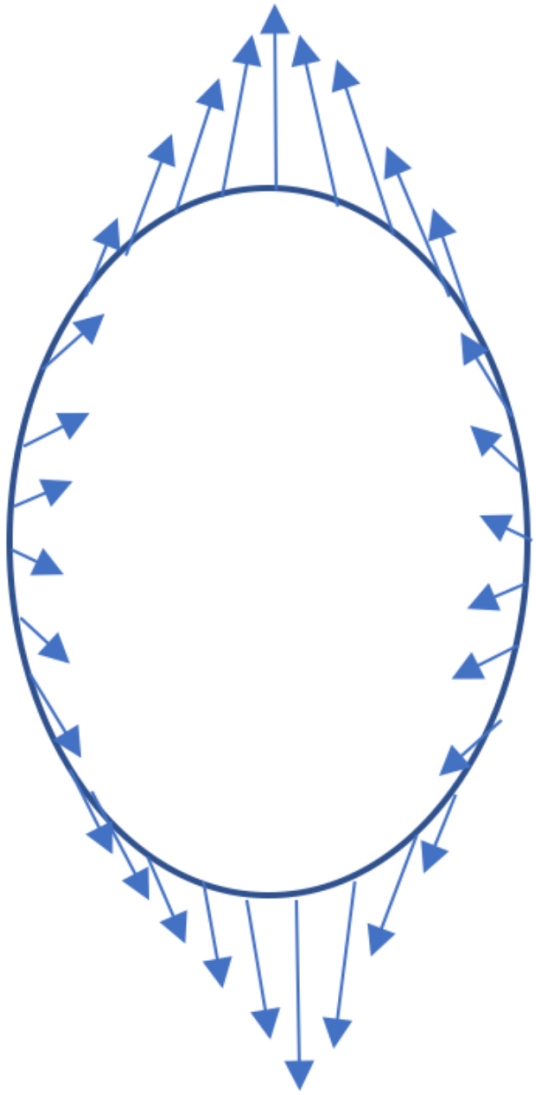


FIG. 1. The tidal forces change the shape of the body into an ellipsoid.

Tidal Locking

The materials that make up the satellite take time to deform and the initial spin rate of the satellite is usually very fast. As a result, as shown in Fig. 2, the semi-major axis of the satellite is misaligned with the line connecting the centers of the two bodies. This misalignment and the non-uniform gravitational field caused by the primary body would cause gravitational torques on the satellite. These gravitational torques are about the center of the satellite.

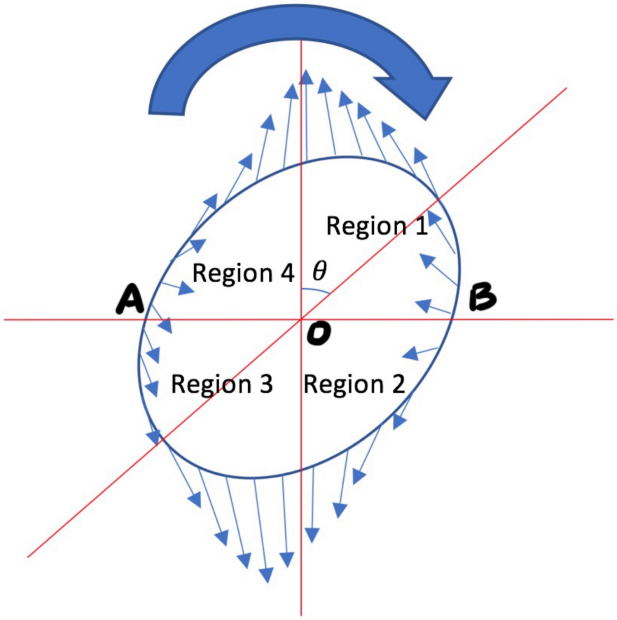


FIG. 2. As shown in the figure, assume the body is spinning clockwise. The body takes time to deform. When the spin rate is very fast, there is a misalignment between the semi-major axis and the line connecting the centers of the two bodies.

The gravitational torque of part i on the satellite is

$$\Gamma_i = \overrightarrow{R}_i \times \left(-2G\frac{Mm_i\Delta r_i}{r^3}\widehat{r}_i\right) \quad (7)$$

Where $\overrightarrow{R}_i$ is the vector pointing from the center of the satellite to part i, $\widehat{r}_i$ is the unit vector pointing from the center of the satellite to the center of the primary body, and m_i is the mass of part i.

Let us assume the satellite is spinning clockwise and its spin period is shorter than its orbital period, as shown in Fig. 2. If taking the direction pointing into the paper as the positive direction, then we have $L_{spin} > 0$.

The total torque can be computed by summing all the torques on every part i. As shown in Fig. 2, the satellite can be divided into four regions and the total torque is

the sum of the torques on these four regions.

$$\Gamma_{tot} = \sum_i \Gamma_i = \sum_i \overrightarrow{R}_i \times \left(-2G\frac{Mm_i\Delta r_i}{r^3}\widehat{r}_i\right) \tag{8}$$

$$\begin{aligned} &= \sum_{i(region\ 1)} \left(-2G\frac{Mm_i\Delta r_i}{r^3}\right)\overrightarrow{R}_i \times \widehat{r}_i \\ &+ \sum_{i(region\ 2)} \left(-2G\frac{Mm_i\Delta r_i}{r^3}\right)\overrightarrow{R}_i \times \widehat{r}_i \\ &+ \sum_{i(region\ 3)} \left(-2G\frac{Mm_i\Delta r_i}{r^3}\right)\overrightarrow{R}_i \times \widehat{r}_i \\ &+ \sum_{i(region\ 4)} \left(-2G\frac{Mm_i\Delta r_i}{r^3}\right)\overrightarrow{R}_i \times \widehat{r}_i \end{aligned} \tag{9}$$

$$= \Gamma_{region\ 1} + \Gamma_{region\ 2} + \Gamma_{region\ 3} + \Gamma_{region\ 4} \tag{10}$$

Because

$$\begin{cases} \Delta r_i > 0, \overrightarrow{R}_i \times \widehat{r}_i > 0, in\ region\ 1 \\ \Delta r_i < 0, \overrightarrow{R}_i \times \widehat{r}_i > 0, in\ region\ 2 \\ \Delta r_i < 0, \overrightarrow{R}_i \times \widehat{r}_i < 0, in\ region\ 3 \\ \Delta r_i > 0, \overrightarrow{R}_i \times \widehat{r}_i < 0, in\ region\ 4 \end{cases} \tag{11}$$

it is obvious that

$$\begin{cases} \Gamma_{region\ 1} = \sum_{i(region\ 1)} \left(-2G\frac{Mm_i\Delta r_i}{r^3}\right)\overrightarrow{R}_i \times \widehat{r}_i < 0 \\ \Gamma_{region\ 2} = \sum_{i(region\ 2)} \left(-2G\frac{Mm_i\Delta r_i}{r^3}\right)\overrightarrow{R}_i \times \widehat{r}_i > 0 \\ \Gamma_{region\ 3} = \sum_{i(region\ 3)} \left(-2G\frac{Mm_i\Delta r_i}{r^3}\right)\overrightarrow{R}_i \times \widehat{r}_i < 0 \\ \Gamma_{region\ 4} = \sum_{i(region\ 4)} \left(-2G\frac{Mm_i\Delta r_i}{r^3}\right)\overrightarrow{R}_i \times \widehat{r}_i > 0 \end{cases} \tag{12}$$

As shown in Fig. 3, when the spin period is shorter than the orbital period, the volumes of region 1 and region 3 are greater than the volumes of region 2 and region 4.

Therefore, we have

$$\begin{cases} |\Gamma_{region\ 1}| > |\Gamma_{region\ 4}| \\ |\Gamma_{region\ 3}| > |\Gamma_{region\ 2}| \end{cases} \tag{13}$$

Therefore,

$$\Gamma_{tot} = \Gamma_{region\ 1} + \Gamma_{region\ 2} + \Gamma_{region\ 3} + \Gamma_{region\ 4} < 0 \tag{14}$$

Because $L_{spin} > 0$, the total gravitational torque reduces the spin rate of the satellite.

As the spin rate of the satellite is decreasing, the angle θ between the semi-major axis and the line connecting the centers of the two bodies decreases. As a result, the difference in volume between region 1 and region 4 as well as the difference in volume between region 3 and region 2 are decreasing. It causes Γ_{tot} to decrease until θ is equal to 0, i.e. when the spin period is equal to the orbital period. When θ is 0, the satellite is tidally locked. Similar analysis can show that the spin rate of the satellite would increase if its initial spin period is longer than its orbital period until the two periods become equal.

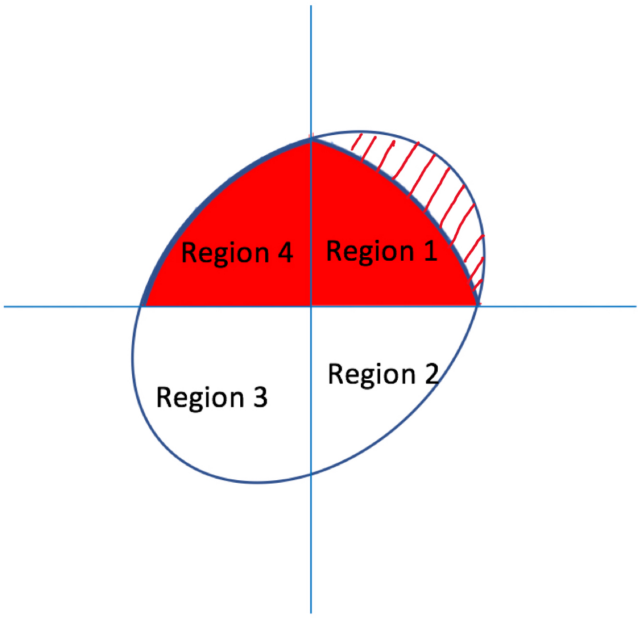


FIG. 3. The volumes of the parts in red in region 1 and region 4 are the same. The gravitational torques acting on them have the same size but opposite directions. The parts in shadow in region 1 indicate there is extra torque acting on region 1. The same logic applies to region 2 and region 3.

The time it takes for a planet to be tidally locked

The tidal locking time can be estimated using the following equation [5]

$$t_{lock} \approx \frac{\omega r^6 IQ}{3GM^2 k_2 R_s^5} \tag{15}$$

Where ω is the initial spin angular velocity of the satellite, r is the length of the semi-major axis of the satellite's orbit around the primary body, I is the moment of inertia of the satellite, Q is the dissipation function of the satellite, G is the gravitational constant, M is the mass of the primary body, k_2 is the tidal Love number of the satellite, and R_s is the mean radius of the satellite.

The tidal Love number of the satellite can be estimated using the following equation [5]

$$k_2 \approx \frac{1.5}{1 + \frac{19\mu}{2\rho g R_s}} \tag{16}$$

Where μ is the rigidity of the satellite, ρ is the density

of the satellite, and g is the gravitational acceleration on the surface of the satellite.

The effect of tidal locking on the orbit of the satellite

Because there is no external force, the total angular momentum of the two body system is conserved. When the satellite's spin angular momentum is decreasing, the satellite's orbital angular momentum has to increase. The total energy of the satellite is [8]

$$E = \frac{1}{2}m\dot{r}^2 + \frac{L_{orbit}^2}{2mr^2} - G\frac{Mm}{r} \tag{17}$$

The effective potential energy is

$$V_{eff} = \frac{L_{orbit}^2}{2mr^2} - G\frac{Mm}{r} \tag{18}$$

Where L_{orbit} is the orbital angular momentum of the satellite.

As shown in Fig. 4, if the orbit of the satellite is assumed to be circular, then the radius of its orbit would increase when its L_{orbit} increases. If the orbit is not circular, then the length of the orbit's semi-major axis would increase.

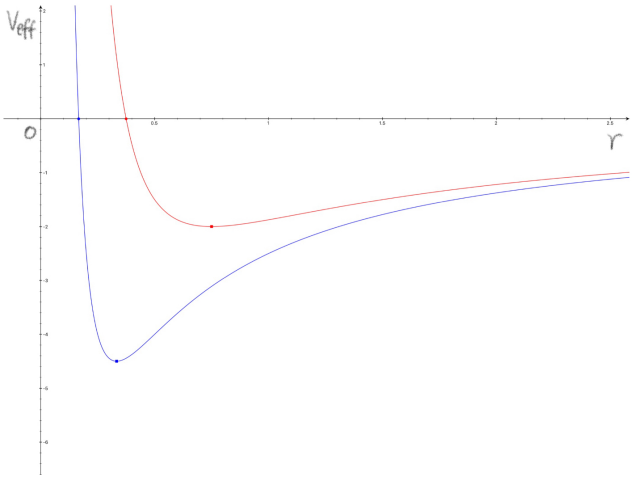


FIG. 4. These are the plots of V_{eff} vs. r for two different L_{orbit}. The curve in red corresponds to the larger L_{orbit} and the curve in blue corresponds to the smaller L_{orbit}. The small red dot and blue dot on the two curves represent the cases where the orbits are circular. According to these plots, it is clear that when L_{orbit} increases, the radius of the orbit tends to increase.

METHODS

Tidal locking time

According to Eqn. (15), the tidal locking time depends on a lot of factors. It is better to simplify Eqn. (15) before doing estimations.

It is reasonable to assume the satellite is spherical when estimating its moment of inertia. Then the satellite's moment of inertia is $\frac{2}{5}mR_s^2$. It is also reasonable to assume the initial spin period is 12 hours, which means $\omega \approx 1.5 \times 10^{-4}\ rad/s$. For many bodies, the dissipation function Q can be estimated as about 100 [9]. Because the tidal Love number k_2 is usually very small, it can be estimated as about $\frac{3\rho g R_s}{20\mu}$ [10].

According to these estimations:

$$t_{lock} \approx \left(\frac{1.5 \times 10^{-4} \times 0.4 m R_s^2 \times 100 r^6}{3 \times 6.67 \times 10^{-11} M^2 \frac{3\rho g R_s}{20\mu} R_s^5} \right) sec \tag{19}$$

$$t_{lock} \approx \left[\frac{6 \times 10^{-3} r^6 m R_s^2 \mu}{3 \times 10^{-11} M^2 \rho \left(G\frac{m}{R_s^2} \right) R_s^6} \right] sec \tag{20}$$

$$t_{lock} \approx \left[\frac{2 \times 10^{19} r^6 m R_s^2 \mu}{6.67 M^2 \left(\frac{4}{3}\pi R_s^3 \rho \right) m R_s \times \frac{3}{4\pi}} \right] sec \tag{21}$$

$$t_{lock} \approx \left(\frac{2 \times 10^{19} r^6 m R_s^2 \mu}{1.6 M^2 m^2 R_s} \right) sec \tag{22}$$

$$t_{lock} \approx \left(\frac{r^6 R_s \mu}{m M^2} \times 1.3 \times 10^{19} \right) sec \tag{23}$$

Then we have

$$t_{lock} \approx \left(\frac{r^6 R_s \mu}{m M^2} \times 4 \times 10^{11} \right) years \tag{24}$$

Where the rigidity parameter μ can be estimated as about 3×10^{10} for most rocky bodies [10].

The effect of tidal locking on the orbit of the Earth-Moon system

According to Eqn. (15), the tidal locking time of a satellite is proportional to the initial spin angular velocity, i.e. $\frac{dt_{lock}}{d\omega} = constant$. As a result, the spin angular velocity ω would decrease at a constant rate while the satellite is being tidally locked. This constant rate can be calculated using the following equation:

$$\frac{d\omega_{spin}}{dt} \approx -\frac{\omega_{spin}}{t} \approx -\frac{3GM^2 k_2 R_s^5}{r^6 IQ} \tag{25}$$

Then the rate of change of the earth's spin period is

$$\frac{dT_{spin}}{dt} = 2\pi \left(-\frac{1}{\omega_{spin}^2}\right) \frac{d\omega_{spin}}{dt} \tag{26}$$

For simplicity, let us assume the orbit is always circular. To find the radius r_c when the orbit is circular, we use

$$0 = \frac{dV_{eff}}{dr}\bigg|_{r_c} = -\frac{L_{orbit}^2}{mr_c^3} + G\frac{Mm}{r_c^2} \tag{27}$$

Then we have

$$r_c = \frac{L_{orbit}^2}{GMm^2} \tag{28}$$

and

$$\frac{dr_c}{dt} = \frac{2L_{orbit}}{GMm^2} \cdot \frac{dL_{orbit}}{dt} \tag{29}$$

$$\frac{dr_c}{dt} = \frac{2r_c^2\omega_{orbit}}{GMm} \cdot \left(-\frac{dL_{spin}}{dt}\right) \tag{30}$$

$$\frac{dr_c}{dt} = \frac{2r_c^2\omega_{orbit}I}{GMm} \cdot \left(-\frac{d\omega_{spin}}{dt}\right) \tag{31}$$

Where M is the mass of the moon, m is the mass of the earth, and I is the moment of inertia of the earth.

RESULTS

The estimation of tidal locking time

The estimations of the tidal locking time using Eqn. (24) for several two-body systems are listed in Table I.

TABLE I. The estimated tidal locking time for some astronomical bodies. The data used to do the estimations and the ages of the systems come from the NASA database [11].

	Estimated time[a]	Age of the system[a]
Moon(Earth)[b]	2.5×10^7	4.51×10^9
Earth(Moon)	7.5×10^9	4.51×10^9
Earth(Sun)	3.6×10^{10}	4.54×10^9
Mars I(Mars)	2.0×10^5	at least 3.5×10^9
Mars II(Mars)	2.0×10^7	at least 1.0×10^9
Mercury(Sun)	8.4×10^8	at least 4.0×10^9

[a] All numbers are in units of years.
[b] The names before the parentheses are the satellites' names, the names in the parentheses are the primary bodies' names.

The effect of tidal locking on the orbit of the Earth-Moon system

For the Earth-Moon system, if we consider the tidal locking of the earth by the moon, then the parameters in Eqn. (25) can be estimated as

$$\begin{aligned} I &\approx \left[0.4 \times 6.0 \times 10^{24} \times \left(6.4 \times 10^6\right)^2\right] kg \cdot m^2 \\ &\approx 9.8 \times 10^{37} kg \cdot m^2 \end{aligned} \tag{32}$$

$$\begin{aligned} k_2 &\approx \frac{1.5}{1 + \dfrac{19 \times 3 \times 10^{10}}{2 \times (5.5 \times 10^3) \times 10 \times (6.4 \times 10^6)}} \\ &\approx 0.8 \end{aligned} \tag{33}$$

$$Q \approx 100 \tag{34}$$

Insert corresponding numbers into Eqn. (25), then we get

$$\begin{aligned} \frac{d\omega_{spin}}{dt} &\approx -3.1 \times 10^{-22}\ rad/s^2 \\ &\approx -9.8 \times 10^{-15}\ (rad/s)/year \end{aligned} \tag{35}$$

Then the rate of change of the earth's spin period is

$$\begin{aligned} \frac{dT_{spin}}{dt} &\approx 2\pi \times \frac{1}{(7 \times 10^{-5})^2} \times 3.1 \times 10^{-22}\ s/s \\ &\approx 4.0 \times 10^{-13}\ s/s \\ &\approx 1.3\ ms/century \end{aligned} \tag{36}$$

By inserting the corresponding numbers into Eqn. (31), we get the rate of change of the distance between the earth and the moon.

$$\frac{dr_c}{dt} \approx 9.0 \times 10^{-10}\ m/s \approx 2.8\ cm/year \tag{37}$$

DISCUSSION

According to the results shown in Table I, the tidal locking times of the moon, Mars I, Mars II, and Mercury are shorter than their ages. Therefore, these bodies should have already been tidally locked. According to the observation, it is true that all of them have been tidally locked [1, 12, 13]. This agrees with my results. However, Mercury is tidally locked with a 3:2 spin-orbit resonance, i.e. Mercury's orbital period is 1.5 times as long as its spin period [13]. Mercury's spin period is not equal to its orbital period after it has been tidally locked. This disagreement between the observation and our theory is because in this paper we have assumed all orbits to be circular but Mercury's orbit is an ellipse. The detailed analysis of why an elliptical orbit can cause a tidal locking with a 3:2 spin-orbit resonance has been presented in other papers [13].

According to Table I, the time for the earth to be tidally locked by the moon and for the earth to be tidally locked by the sun is much longer than the age of the corresponding two-body systems. This means earth has not been tidally locked by the moon or the sun and it still requires a very long time (on the order of billion years)

for the earth to be tidally locked by the moon or the sun. Obviously, this also agrees with the observations. These agreements of theoretical estimations and observations indicate that the theory about tidal locking is reliable.

According to the observation, the earth's period of spin is increasing at a rate of 2.3 ms/century [6]. According to my estimation, the earth's period of spin is increasing at a rate of about 1.3 ms/century. These results generally agree because they are on the same order.

According to the observation, the distance between the moon and the earth is increasing by about 3.8 cm/year [7]. This is also very close to my estimation, which is about 2.8 cm/year. Even though I made many assumptions which are not true when doing the estimation, e.g. the orbit is always circular and all bodies share the same $Q = 100$, the estimation based on the theory agrees pretty well with the observations. The differences between the estimated values and the observed values are probably due to the simplifications that I made in the model. These agreements again support our theory about tidal locking.

CONCLUSION

The model in this paper is very simple. All orbits are assumed to be circular. The estimations of the tidal dissipation function Q and the tidal Love number k_2 are not accurate. Nevertheless, the theoretical estimations generally agree with observations. This indicates this model is successful. It is possible to make the estimations more accurate by considering the true elliptical orbits of the bodies and including more accurate Q and k_2.

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